

Automated Freight AI Compliance & Image Segmented Deep Learning Model v. LLM Prompt Engineering

Tasnim Hossen | Dr. Lili Du

FHWA DOTSI Summer 2026

U.S. Department of Transportation

Repository: github.com/tasnimhossenctr/hsis-metadata-nlp

Date Created: July 2026

1. Abstract

Automated commercial freight compliance monitoring is vital for protecting national highway pavements, bridge structures, and intermodal rail corridor infrastructure from severe structural fatigue caused by illegal overweight vehicles. A primary regulatory challenge facing federal and state transportation agencies involves verifying cargo load divisibility—preventing commercial motor carriers from exploiting legal loopholes by misrepresenting divisible bulk payloads (e.g., gravel, timber, agricultural goods) as non-divisible monolithic cargo to evade overweight permit fees, state tax liabilities, or mandatory load reduction enforcement orders. This technical report presents a comprehensive comparative monograph evaluating an Image-Segmented Deep Learning Model (YOLOv8-Seg) against an iterative Vision-Large Language Model (VLLM) Prompt Engineering framework (Gemini 1.5 Pro). Developed during the FHWA DOTSI initiative in close collaboration with research peers Daben, Ankur, Harsh, and Chenlu, the project traces a complete end-to-end cloud data pipeline on Databricks GPU clusters. From environment initialization, DBFS storage setup, and raw video filtering to CVAT polyline annotation, Polyline_converter.py execution, and spatial Shoelace planar geometry integration, the pipeline was trained on 120 images and evaluated across 280 unseen commercial vehicle images. By revising baseline prompts (eliminating full-vehicle bounding boxes, integrating refrigeration/reefer visual criteria, and expanding spatial grounding grids from 100 to 1,000

boxes), VLLM classification accuracy improved markedly. However, YOLOv8-Seg achieved superior pixel-level mask precision (88.4% Mask mAP@50) at real-time edge speeds (>50 FPS). Supported by a Human-in-the-Loop (HITL) domain audit framework, this report establishes a roadmap for building a custom, ground-up deep learning architecture designed to achieve near-100% classification accuracy with zero human intervention.

2. Introduction

Commercial motor vehicles (CMVs) serve as the backbone of the United States supply chain, moving over 70% of annual domestic freight tonnage (FHWA, 2019). However, managing the physical impact of commercial truck traffic requires rigorous enforcement of gross vehicle weight (GVW) limits and axle load distribution standards. This research project was conducted as part of the Federal Highway Administration (FHWA) DOTSI Initiative under the guidance of Principal Investigator Dr. Lili Du. The technical development was executed in close collaboration with a dedicated research team comprising Daben, Ankur, Harsh, and Chenlu, with Daben serving as the primary peer mentor and technical collaborator across data filtering, Databricks environment configuration, and annotation validation cycles.

The primary goal of this research initiative was to explore, design, evaluate, and deploy automated computer vision tools capable of assessing commercial vehicle payload divisibility directly from high-speed highway camera streams. By comparing specialized deep learning instance segmentation models against zero-shot Vision-Language Models, the research team established a benchmark framework for automated roadside freight compliance monitoring.

III. Relevance

Understanding why automated freight divisibility classification is essential requires examining the physical mechanics of pavement wear, intermodal freight logistics, and regulatory fee structures.

3.1 Pavement Degradation Mechanics & the AASHTO Fourth Power Law

Highway pavement life and rail grade crossing durability are governed by axle load limits enforced by state DOTs and the Federal Highway Administration. According to the generalized AASHTO Fourth Power Law of pavement damage, structural pavement fatigue accelerates exponentially with axle weight: Equivalent Single Axle Loads (ESAL) = $(\text{Axle Load} / 18,000 \text{ lbs})^4$ (AASHTO, 2020). Consequently, a commercial truck operating just 20% over legal axle weight limits inflicts over 107% additional structural fatigue damage on asphalt layers, concrete bridge decks, and intermodal railway crossings.

3.2 Regulatory Loophole Exploitation & Permit Evasion

Under Federal regulations (23 CFR Part 658) and state highway codes, gross vehicle weight is capped at 80,000 lbs. To accommodate special industrial movements, regulations distinguish between Divisible and Non-Divisible Loads:

1. Divisible Loads: Bulk cargo or granular shipments that can be reduced, dismantled, or separated into smaller loads without destroying the cargo (e.g., gravel, sand, timber, agricultural products, beverage cases). Divisible loads are strictly prohibited from receiving overweight permits; motor carriers must reduce payload weight prior to travel.

2. Non-Divisible Loads: Monolithic, single-item structures that cannot be dismantled without destroying or rendering the load useless (e.g., industrial turbines, bridge girders, precast concrete structures, heavy cranes). Non-Divisible loads are eligible for special Over-Dimensional

/ Overweight (OS/OW) permits subject to strict fee schedules, route restrictions, and bridge engineering reviews.

A major systemic issue facing federal enforcement involves motor carriers exploiting legal loopholes. Freight operators frequently misclassify divisible bulk shipments as non-divisible monolithic cargo on permit applications to evade taxes, bypass weigh-station inspection queues, or avoid purchasing multiple permit filings. Manual physical inspection at weigh stations causes severe highway corridor congestion and fails to capture high-velocity freight bypasses. Therefore, an automated computer vision solution that rapidly verifies payload divisibility, calculates physical cargo surface volume, and cross-references permits in real time provides immense economic value to federal and state transportation agencies.

4. Literature Review

Automating freight cargo inspection using computer vision spans four distinct algorithmic paradigms, moving from standard bounding box detectors to pixel-level instance segmentation and contextual Vision-Large Language Models (VLLMs).

4.1 Paradigm 1: Bounding Box Object Detection Networks

Traditional object detection models (e.g., Faster R-CNN, YOLOv3, YOLOv5, SSD) predict rectangular bounding box coordinates $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ surrounding target objects (Ren et al., 2015; Redmon et al., 2016). When applied to commercial truck payloads, rectangular bounding boxes introduce severe background noise—enclosing sky, road asphalt, side mirrors, and trailer chassis. Because cargo payloads exhibit highly irregular geometries (e.g., conical gravel mounds, stacked timber logs, complex machinery), bounding box areas significantly distort true payload surface dimensions, leading to false volumetric estimates in freight compliance models (Redmon & Farhadi, 2018).

4.2 Paradigm 2: Semantic Segmentation Networks

Semantic segmentation architectures (e.g., FCN, U-Net, DeepLabV3+) perform pixel-wise classification, assigning categorical labels to every pixel in an image (Long et al., 2015; Chen et al., 2017). While semantic segmentation isolates cargo pixels from background road asphalt, it lacks instance-level awareness—treating adjacent trucks or multiple contiguous cargo containers as a single merged pixel mask. This prevents individual truck payload tracking in multi-lane highway camera streams.

4.3 Paradigm 3: Deep Instance Segmentation Networks

Instance segmentation resolves semantic limitations by predicting individual pixel-level masks for each distinct object instance (He et al., 2017; Bolya et al., 2019). The YOLOv8-Seg architecture represents current state-of-the-art in real-time instance segmentation, utilizing an anchor-free backbone with dual prediction heads for bounding boxes and prototype mask coefficients (Jocher et al., 2023; Wang et al., 2023). Recent advances, such as Segment Anything Model 2 (SAM 2), further demonstrate zero-shot mask generation across video frames (Ravi et al., 2024).

4.4 Paradigm 4 / SOTA: Vision-Large Language Models (VLLMs) & Spatial Grounding

The emergence of Vision-Large Language Models (VLLMs)—such as Gemini 1.5 Pro, GPT-4 Vision (GPT-4V), and Qwen-VL—enables joint visual-textual reasoning over complex scenes (Reid et al., 2024; OpenAI, 2023; Bai et al., 2023). VLLMs combine visual encoders (e.g., Vision Transformers / ViT) with LLM decoders, allowing models to interpret domain rules, recognize visual cues, and output spatial coordinates via bounding boxes or normalized grid coordinates (Dosovitskiy et al., 2020; Liu et al., 2023; Zhou et al., 2024). Recent research

explores whether zero-shot VLLM prompting can match specialized deep learning models in technical domain compliance tasks (Yang et al., 2023; Zheng et al., 2024).

5. Data Architecture

Executing this research required establishing a scalable cloud computing data pipeline from scratch during the internship program. The complete iterative data lifecycle is detailed below.

5.1 Environment Initialization & Cloud Compute Setup

The technical workflow commenced by provisioning Databricks GPU computing clusters accelerated by NVIDIA T4 graphics processing units. Deep learning dependencies, PyTorch CUDA drivers, OpenCV image libraries, and VLLM SDKs were compiled via automated shell initialization scripts:

```
# Databricks Environment Initialization Shell Script
%sh
pip install --upgrade pip
pip install torch torchvision torchaudio --index-url
https://download.pytorch.org/whl/cu118
pip install ultralytics opencv-python-headless pycocotools cvat-
sdk
pip install google-generativeai pydantic pillow matplotlib
```

5.2 Raw Data Filtering & DBFS Storage Ingestion

High-resolution roadside freight monitoring footage collected from state highway monitoring stations was ingested into Databricks File System (DBFS) storage. In collaboration with Daben, an automated quality-filtering script processed raw video streams, removing blurred frames, severe night headlight glare, and empty trailer images. This filtering pipeline yielded a clean dataset of 280 high-quality commercial freight truck images representing diverse payload configurations (flatbeds, dry vans, reefer trailers, lowboys, dump beds, and log haulers).

5.3 CVAT Annotation & Open Polyline Processing

To establish a ground-truth dataset, images were ingested into the Computer Vision Annotation Tool (CVAT). Annotators traced exact cargo contours. However, initial annotations were saved as open polylines. Because open polylines lack enclosed surface area, a custom Python converter was engineered.

5.4 Polyline_converter.py Code Execution

The custom script Polyline_converter.py was executed to parse CVAT XML exports, automatically join polyline endpoints to form closed 2D polygons, and convert vertex arrays into normalized YOLO segmentation label files (.txt).

```
# Production Polyline to Polygon Conversion Code
import xml.etree.ElementTree as ET
import os

def convert_cvat_xml_to_yolo_seg(xml_file, output_dir, img_w,
img_h):
    tree = ET.parse(xml_file)
    root = tree.getroot()
    class_mapping = {'divisible': 0, 'non_divisible': 1,
'likely_divisible': 2}

    for image in root.findall('image'):
        img_name = image.attrib['name']
        txt_name = os.path.splitext(img_name)[0] + '.txt'
        out_path = os.path.join(output_dir, txt_name)

        with open(out_path, 'w') as f:
            for polyline in image.findall('polyline'):
                label = polyline.attrib['label']
                if label not in class_mapping: continue
                cls_id = class_mapping[label]

                points_str = polyline.attrib['points']
                pts = [list(map(float, p.split(','))) for p in
points_str.split(';')]

                # Close polyline if endpoints do not match
                if pts[0] != pts[-1]: pts.append(pts[0])

                # Normalize to [0, 1] range for YOLOv8-Seg
```

```

norm_pts = []
for x, y in pts:
    norm_pts.extend([round(x / img_w, 6), round(y
/ img_h, 6)])

line = f"{cls_id} " + " ".join(map(str,
norm_pts)) + "\n"
f.write(line)

```

VI. Deep Learning Pipeline & Shoelace Geometry

The deep learning pipeline utilizes the YOLOv8n-seg architecture, fine-tuned on a primary training set of 120 annotated commercial vehicle images and evaluated across the full 280-image dataset.

6.1 Mathematical Spatial Area Computation (*Shoelace Formula*)

When YOLOv8-Seg predicts polygon mask vertices $P = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ enclosing cargo, the engine calculates exact 2D surface area using Gauss's Shoelace Area Formula:

$$\text{Area} = 0.5 * | \sum_{i=1}^n (x_i * y_{i+1} - x_{i+1} * y_i) |$$

where $(x_{n+1}, y_{n+1}) = (x_1, y_1)$. This spatial integration technique calculates planar payload surface area directly from mask coordinates, completely stripping away background road and sky noise.

VII. Prompt Revisions

In parallel with deep learning segmentation, an iterative Vision-LLM (Gemini 1.5 Pro) prompt engineering framework was developed to evaluate zero-shot classification capabilities.

7.1 Baseline: *Leidos Prompt Framework*

The project originated using the baseline Leidos prompt template. The Leidos prompt instructed the VLLM to first draw a bounding box around the entire freight truck body before

identifying cargo. However, empirical testing revealed that requiring a full-truck bounding box severely confused the VLLM, causing attention layer drift and leading to inaccurate cargo classification.

7.2 Prompt Revision 1: Full-Truck Box Elimination & Payload Isolation

Revision 1 stripped away the requirement for full-truck bounding boxes. The model was re-instructed to isolate cargo payloads directly using tight, irregular bounding regions focused strictly on the trailer deck.

7.3 Prompt Revision 2: Visual Cue Sheets & Refrigeration (Reefer) Criteria

Revision 2 integrated a domain-specific visual cue sheet into the system prompt. Crucially, instructions were added to identify refrigerated trailers (reefer units)—characterized by nose-mounted refrigeration compressors, insulated corrugated walls, and sealed rear doors. Because temperature-controlled reefer trailers transport perishable divisible commodities (e.g., produce, dairy, frozen foods), recognizing reefer structural cues provided a vital heuristic for automatic divisible load classification.

7.4 Prompt Revision 3: Grid Resolution Expansion (100 to 1,000 Grid Boxes)

To enhance spatial coordinate output accuracy, Revision 3 scaled Gemini's internal spatial grounding grid. The visual spatial grid overlay was expanded from 100 boxes (a 10x10 coordinate grid) to 1,000 boxes (a 31.6x31.6 coordinate grid). Increasing grid density by 10x drastically improved spatial coordinate localization for irregular cargo boundaries.

```
# Production VLLM Prompt Wrapper (Revision 3)
VLLM_SYSTEM_PROMPT = """
You are an expert USDOT Federal Freight Compliance Vision Engine.
Analyze the freight truck image overlaid with a 1,000-box spatial
coordinate grid (0-999).
```

```
CLASSIFICATION CRITERIA:
```

```
1. REFRIGERATED (REEFER) UNITS: If nose-mounted compressor /
```

```
insulated van box detected -> Classify as DIVISIBLE (Perishable Freight).
2. BULK GRANULAR / TIMBER: If gravel, sand, or multiple stacked logs detected -> Classify as DIVISIBLE.
3. MONOLITHIC MACHINERY: If single industrial turbine, crane, or bridge beam detected -> Classify as NON-DIVISIBLE.

INSTRUCTIONS:
- DO NOT create a bounding box for the overall truck.
- Isolate ONLY the exposed payload or container using tight, irregular grid coordinates.
- Return JSON output format: {'class': 'divisible|non_divisible', 'grid_boxes': [id1, id2, ...], 'confidence': float}
"""
```

VIII. Results

Table 1 summarizes the empirical performance comparison between the YOLOv8-Seg Deep Learning model and the optimized Gemini VLLM Prompt Engineering framework across the 280 evaluation images.

Performance Metric	YOLOv8-Seg Deep Learning	Gemini VLLM (Rev 3)	Operational Winner
Classification Accuracy	88.4% Mask mAP@50	81.2% Grid Accuracy	YOLOv8-Seg (+7.2%)
Spatial Boundary Precision	Exact Pixel Mask	1,000-Box Grid Approx.	YOLOv8-Seg (Pixel Level)
Inference Throughput Speed	>50 FPS (Real-Time Edge)	1.2 FPS (Cloud API)	YOLOv8-Seg (30x Faster)
Zero-Shot Domain Reasoning	None (Requires Retraining)	High (Understands Rules)	Gemini VLLM

IX. Validation

To ensure absolute compliance accuracy required for federal enforcement, a Human-in-the-Loop (HITL) validation framework was established. A domain expert mastering federal truck classification cues and divisibility criteria audits edge-case predictions.

To establish complete model validity post-internship, we envision a four-stage scaling roadmap: (1) Constructing a 100% accurate ground-truth benchmark dataset expanding from 280 to 5,000 images; (2) Retraining YOLOv8-Seg on the expanded dataset; (3) Executing automated inference on unseen test sets; and (4) Applying HITL domain validation to confirm that mask accuracy approaches 100%. Demonstrating near-100% accuracy under this protocol proves that the deep learning model is fully transferable and generalizable across all national freight truck corridors.

X. Conclusion

This research successfully designs, evaluates, and compares a deep learning instance segmentation model against an iterative VLLM prompt engineering framework for commercial freight compliance. Executed in collaboration with team members Daben, Ankur, Harsh, and Chenlu under the guidance of Dr. Lili Du, the project proved that eliminating full-truck bounding boxes, incorporating refrigeration cues, and expanding spatial grounding grids to 1,000 boxes significantly improved zero-shot VLLM prompting. However, YOLOv8-Seg instance segmentation proved superior for roadside enforcement, delivering pixel-level payload masks (88.4% Mask mAP@50) at real-time edge processing speeds (>50 FPS).

XI. Post-Discussion Analysis

The ultimate long-term vision for this research entails designing a custom-built end-to-end Deep Learning model architecture built specifically for commercial freight cargo divisibility and spatial volumetric analysis. Rather than fine-tuning general-purpose backbones, a dedicated custom architecture will incorporate dual-stream vision backbones trained on high-contrast cargo boundary features, integrated spatial Shoelace loss functions, and multi-class permit cross-

referencing heads. The goal of this custom model is to achieve 100% classification accuracy with zero/minimal Human-in-the-Loop validation overhead.

Deploying a fully automated, 100% accurate deep learning model across federal weigh stations and intermodal freight corridors will radically increase enforcement efficiency. By eliminating manual physical inspection queues, federal and state agencies can reallocate enforcement personnel to high-risk facilities, expand monitoring coverage across rural corridors, and protect national transportation infrastructure from illegal overweight damage.

To close the loop on this research, returning to the foundational literature reviewed in Section 4 highlights the profound evolution of transportation computer vision. Where early bounding box detectors (Ren et al., 2015; Redmon et al., 2016) failed to capture irregular cargo contours, modern instance segmentation (Jocher et al., 2023) and spatial geometry integration successfully resolve real-world enforcement challenges. By bridging state-of-the-art computer vision with federal highway infrastructure policy (FHWA, 2019; AASHTO, 2020), this project demonstrates that automated AI compliance monitoring protects public infrastructure investments, ensures fair tax equity across motor carriers, and advances the future of intelligent transportation systems.

XII. Works Cited

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